

FUQ-72XX Series FM Stereo Broadcast Transmitter 1x RF Output FPGA Digital Modulation LDMOS Amplifier $\leq 0.2\%$ Distortion GB/T4311-2000 Standard High Fidelity Transmission for FM Radio and Emergency Alert Systems



Product overview

The FUQ-72XX Series is a professional FM stereo broadcast transmitter with 1x RF output and LDMOS amplifier technology. It processes audio signals via FPGA-based digital modulation (DAC) and delivers 10W-10kW adjustable power with $\leq 0.2\%$ distortion. The rack-mountable unit features air-cooled operation and complies with GB/T4311-2000/GY/T169-1 standards. Ideal for FM radio stations, emergency broadcast systems, and community radio networks requiring high-fidelity transmission.

Key feature

Universal PSU: Supports AC100–264V input and -20°C to $+50^{\circ}\text{C}$ operation for harsh environments.

Technical Specifications

S/No.	Item	Item	Unit	Technical Index	Technical Index	Remark
1	Output Power	Output Power	W	500	500	
2	Frequency	Frequency	MHz	$1\text{kW} \leq$	65.5~87; 87~108	
2	Frequency	Frequency	MHz	$1\text{kW} \blacksquare$	$87 \blacksquare 108$	
3	Frequency deviation	Frequency deviation	Hz	≤ 500	≤ 500	
4	Sparious Radiation	Sparious Radiation	dB	≤ -70	≤ -70	

5	parasitic amplitude modulation noise	parasitic amplitude modulation noise	dB	-60	-60	
6	Pilot frequency deviation	Pilot frequency deviation	Hz	±1	±1	
7	S signal 38kHz Residual component	S signal 38kHz Residual component	dB	■-40	■-40	
8	Deviation (100% modulation)	Deviation (100% modulation)	kHz	±75	±75	
9	pre-emphasis	pre-emphasis	μs	50	50	
10	Distortion (100% modulation)	L	■	■0.2	■0.2	
10	Distortion (100% modulation)	R	■	■0.2	■0.2	
11	Frequency response (without emphasis, de-emphasis)	L	dB	±0.2	±0.2	
11	Frequency response (without emphasis, de-emphasis)	R	dB	±0.2	±0.2	
12	Frequency response (with emphasis, de-emphasis)	L	dB	±0.2	±0.2	

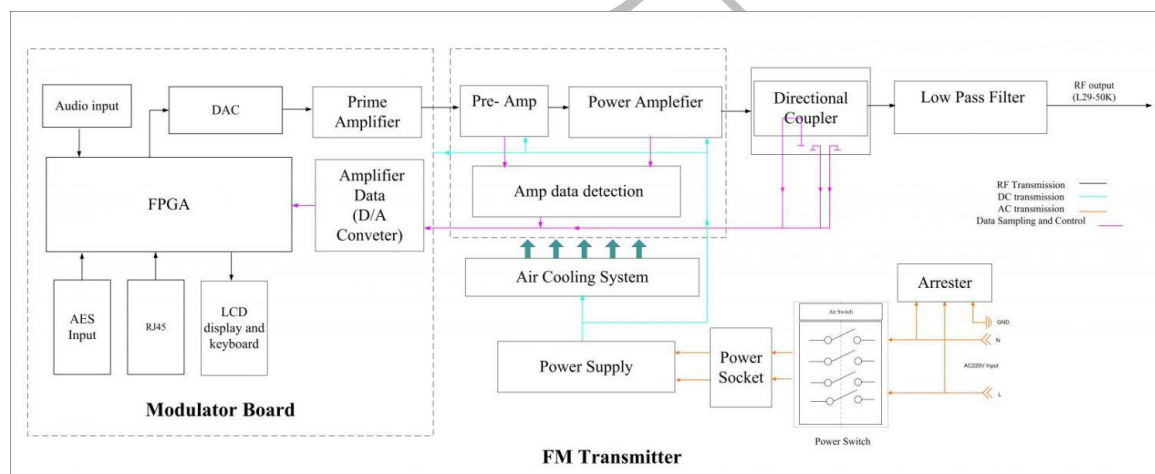
12	Frequency response (with emphasis, de-emphasises)	R	dB	±0.2	±0.2	
13	SNR (100% modulation)	L	dB	≥75	≥75	
13	SNR (100% modulation)	R	dB	≥75	≥75	
14	L/R Separation	L R	dB	■70	■70	
14	L/R Separation	R L	dB	■70	■70	
15	L/R level difference	L/R level difference	dB	■0.1	■0.1	

Specifications and Environment Conditions

Item	Item	Technical Requirements	Technical Requirements	Remark
Output and Input Specifications	RDS input	BNC interface(57KHz)	BNC interface(57KHz)	
Output and Input Specifications	Input interface	XLR, AES/EBU(Optional)	XLR, AES/EBU(Optional)	
Output and Input Specifications	Output interface	L29-50K	L29-50K	
Environment condition	Working temperature	-20■+50■	-20■+50■	
Environment condition	Storage temperature	-30■+75■	-30■+75■	
Environment condition	Relatively humidity	■95% (25■ no condensation)	■95% (25■ no condensation)	
Environment condition	Cooling mode	inside forced air cooling	inside forced air cooling	
Environment condition	atm press	86■106kPa	86■106kPa	

Environment condition	power supply	AC, 100■264V/50Hz	AC, 100■264V/50Hz	
Environment condition	machine room	Less dust, shock-free	Less dust, shock-free	
Environment condition	Demission (L×W×H)	10~30W	482×420×44mm	
Environment condition	Demission (L×W×H)	100W	482×420×88mm	
Environment condition	Demission (L×W×H)	300~1kW	482×560×176m m	
Environment condition	Demission (L×W×H)	3kW	600×800×1080m m	
Environment condition	Demission (L×W×H)	5kW	600×1050×1695 mm	
Environment condition	Demission (L×W×H)	10kW	650×1050×1695 mm	

System Principle



Order Information

DFT-7211: 10W	DFT-7212: 100W	DFT-7213: 1kW	DFT-7214: 10kW
DFT-7231: 30W	DFT-7232: 300W	DFT-7233: 3kW	DFT-7214: 10kW
DFT-7251: 50W	DFT-7252: 500W	DFT-7253: 5kW	DFT-7214: 10kW

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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